Global Environmental Picture

Major Global Environmental Concerns

Population Growth

Soil Degradation

Atmospheric Change

Biodiversity Loss

Global Environmental Concerns

Population Growth





Biodiversity Loss

Atmospheric Change





Soil Degradation

Global Environmental Concerns

Population Growth

Population

Total human inhabitants of a specified area, such as a city, country, or continent, at a given time. The study as a discipline is known as demography.

Deals with

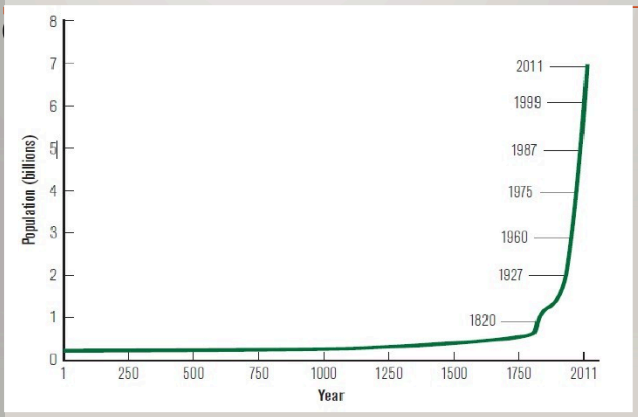
It is concerned with the size, composition, and distribution of populations; their patterns of change over time through births, deaths, and migration; and the determinants and consequences of such changes.

Why do you need to know?

- Knowledge of population studies is important for planning, particularly by governments, in fields such as health, education, housing, social security, employment, and environmental preservation.
- Such studies also provide information needed to formulate government population policies, which seek to modify demographic trends to achieve economic and social objectives.

Population Growth

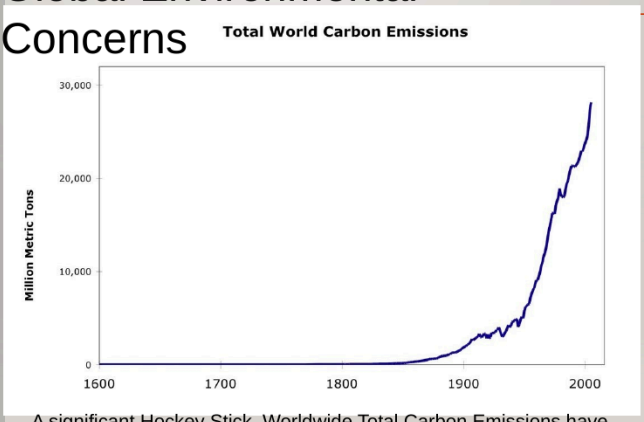
Global Environmental



Don't know what made this chart explode since 1900. Charts show exponential growth and are one of the major causes of Global Warming.

Population Growth

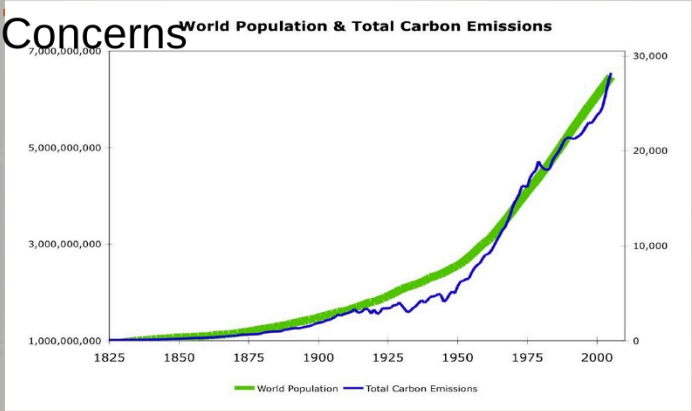
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A significant Hockey Stick. Worldwide Total Carbon Emissions have experienced exponential growth. The chart shows historic Total Annual Worldwide Carbon Emissions since 1600.

Population Growth

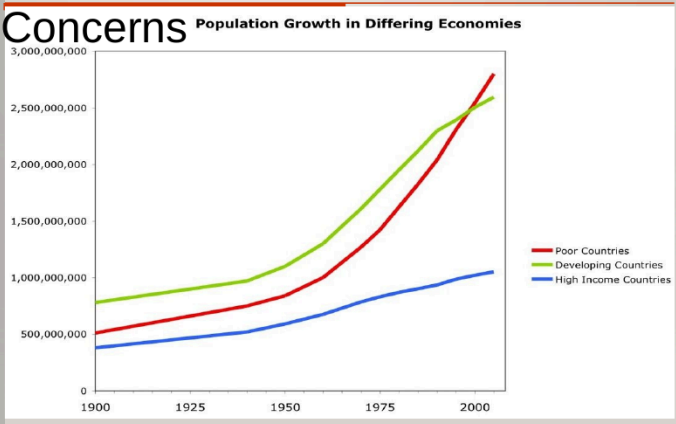
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As Al Gore said, "Yes, they fit." This chart shows how Total Annual Carbon Emissions are directly affected by Population Growth.

Populatio n Growth

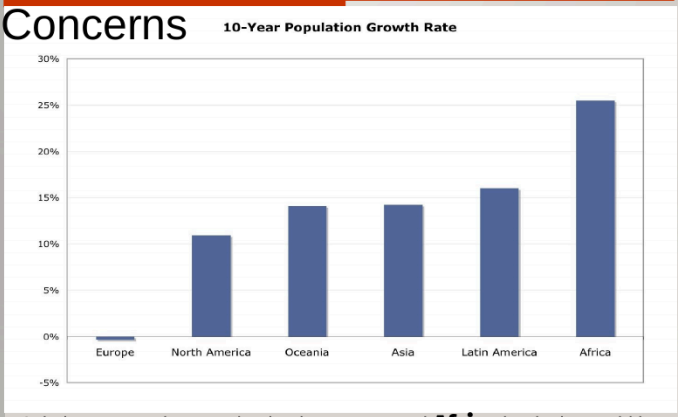
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This chart shows how Population Growth Rates are affected by Income.

Populatio n Growth

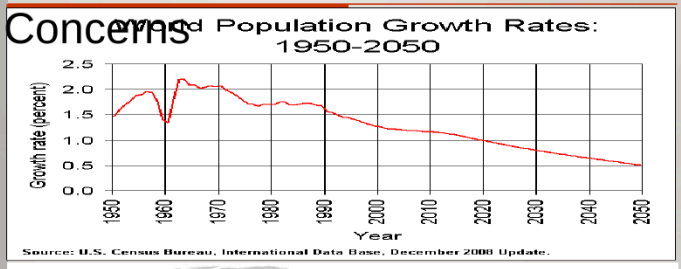
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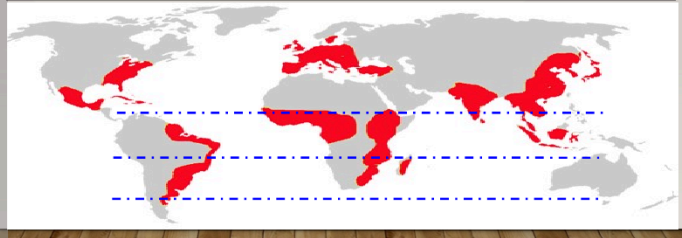


Asia has grown the most in absolute terms, and **Africa** leads the world in its **Rate of Growth**.

Populatio n Growth

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Soil Degradation

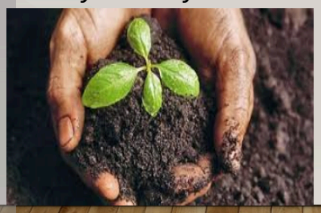
Global Environmental Conc[redacted]

SOIL

… is a marvelous substance, a living resource of astonishing beauty, complexity, and frailty.

It is a complex mixture of weathered mineral materials from rocks, partially decomposed organic molecules, and a host of living organisms.

ecosystem by itself



essential component of the biosphere

Soil Degradation

Global Environmental Concerns

Soil degradation

Based on both biological productivity and our expectations about what the land should be like. Often this is a subjective judgment, and it is difficult to distinguish between human-caused deterioration and natural conditions, like drought. We generally consider the land degraded when the soil is impoverished or eroded, water runs off or is contaminated more then normal, vegetation is diminished, biomass production is decreased, or wildlife diversity diminishes. It entails a decrease in soil's productivity and quality.

Impacts…

On farmlands, lower crop yields…. On ranchlands, fewer livestock can be supported per unit area…On nature reserves, lower biological diversity.

Soil Degradation

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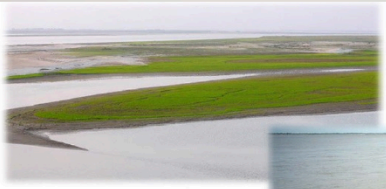
Human-induced soil degradation



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Soil Degradation

Results…!



Water and wind erosion



Char dwellers in Bangladesh

Water induced erosion in Bangladesh

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Soil Degradation

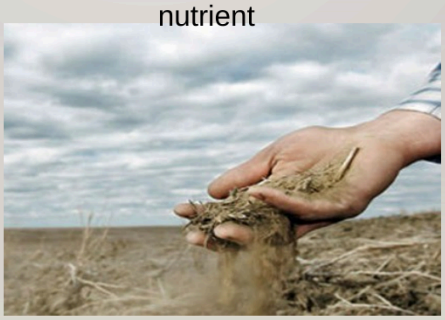
Chemical degradation includes nutrient depletion, salt accumulation, acidification, and pollution



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Soil Degradation

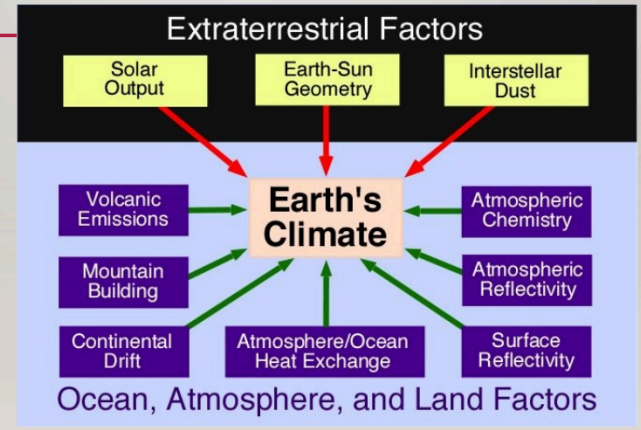
sandy soil…less nutrient



- Overgrazing, monocropping, and the use of agrochemicals, oil contamination from industry, mining, illegal landfills, and poorly managed landfills also contribute to the degradation of soil.
- **The assessment of soil degradation depends on the type of degradation process, the scale of the assessment, and a method that can be based on expert opinion (through questionnaires), remote sensing (satellite imagery), or modelling**

Global Environmental Concerns

Atmospheric Change



EXTRA-TERRESTRIAL FACTORS

- Solar output affects our climate—both directly and indirectly, **by changing the rate of solar heating of the Earth and atmosphere**, and indirectly, by changing cloud-forming processes.
- The Sun-Earth geometry **causes heating differences across the planet**. Different parts of the Earth experience variations in climatic changes due to the change in the Earth's position caused by revolution around the Sun. The four seasons are summer, winter, fall, and spring.

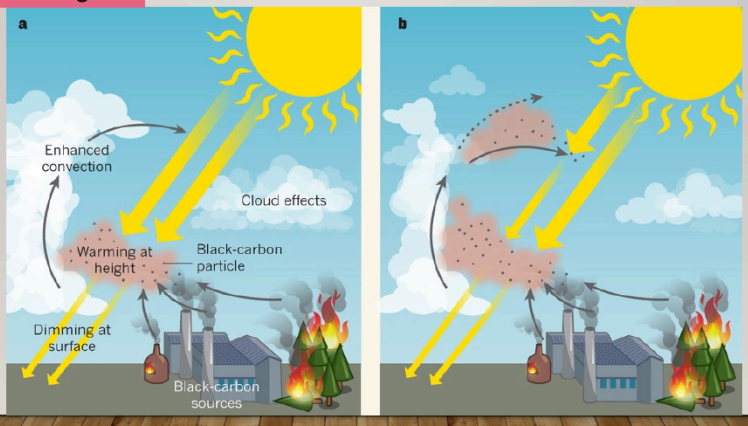
- Continental drift is the hypothesis that the Earth's continents have moved over geologic time relative to each other, thus appearing to have "drifted" across the ocean bed.
- The idea of continental drift has been subsumed into the science of plate tectonics, which studies the movement of the continents as they ride on plates of the Earth's lithosphere

- Atmosphere/ocean heat exchange: the oceans impact weather and climate by heating (and cooling) the lower atmosphere. As **seawater evaporates, the ocean surface cools; when the moisture later condenses into cloud droplets, this heat is released, warming the atmosphere**.
- Mountain building refers to the **geological processes that underlie the formation of mountains.** These processes are associated with large-scale movements of the Earth's crust (tectonic plates). Folding, faulting, volcanic activity, igneous intrusion, and metamorphism can all be parts of the orogenic process. The formation of mountains is not necessarily related to the geological structures found in them.
- Volcanic gases like sulfur dioxide can cause global cooling, while volcanic carbon dioxide, a greenhouse gas, **has the potential to promote global warming.**

Global Environmental Concerns

Atmospheric Change

Human-Induced Factors



WHY IS BIODIVERSITY LOSS A CONCERN?

LIFE HAS EXISTED ON EARTH FOR OVER 3.5 BILLION YEARS. OVER 95% OF THE SPECIES THAT EVER EXISTED HAVE GONE EXTINCT. WHY



Why is biodiversity loss a concern?

Human Well-being
Basic material for a good life
Health
Security
Good social relations
Freedom of choice and action

Ecosystem Services

Provisioning Services
Food, fiber, and fuel
Genetic resources
Biochemical
Fresh Water

Cultural Services
Spiritual and religious values
Knowledge system
Education and inspiration
Recreation and aesthetic values
Sense of place

Supporting Services
Primary production
Provision of habitat
Nutrient Cycling
Soil formation and retention
Production of atmospheric oxygen
Water cycle

Regulating Services
Invasion resistance
Herbivore
Pollination
Seed dispersal
Climate regulation
Disease regulation
Natural hazard protection
Erosion regulation
Water purification
Pest Regulation

Biodiversity
Number
Relative abundance
Composition
Interactions

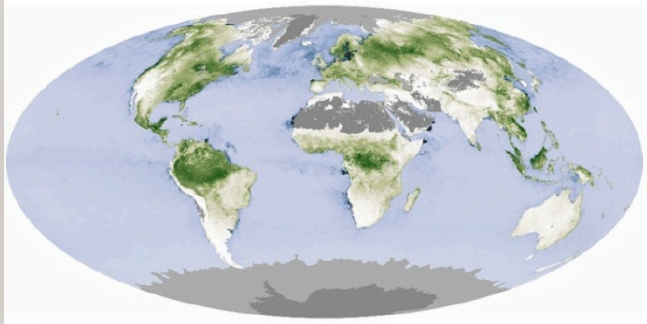
Ecosystem Functions

Millennium ecosystem assessment, 2005

Example: Provisioning services



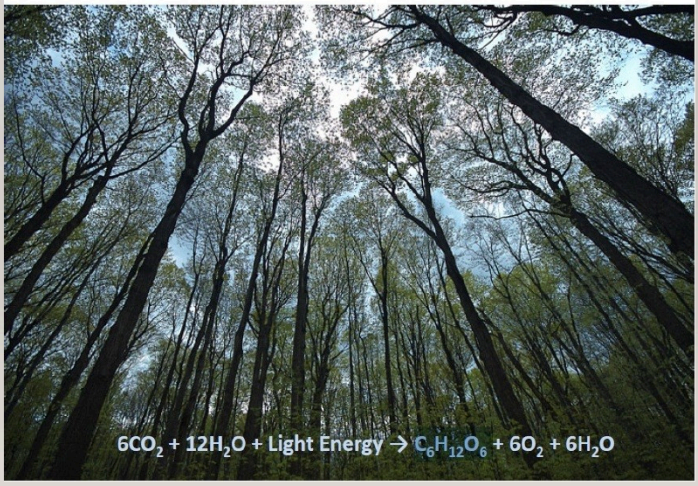
Example: Supporting service



Land and ocean net primary production estimated from spectral data

Source: NASA Earth Observatory

Example: Supporting service



$6CO_2 + 12H_2O + \text{Light Energy} \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$

Example: Regulating services







Human pollination, China